Outils Mathématiques – Calcul Différentiel

CD13

1) $\frac{1}{R}=\frac{1}{R_1}+\frac{1}{R_2} \Leftrightarrow \frac{1}{R}=\frac{R_1+R_2}{R_1 R_2} \Leftrightarrow R=\frac{R_1 R_2}{R_1+R_2}$ donc $R=\frac{3300}{9}\,\Omega \approx 113{,}7931034\,\Omega$

2) Calcul direct

$$\Delta R=\left|\frac{\partial R}{\partial R_1}\right|\Delta R_1+\left|\frac{\partial R}{\partial R_2}\right|\Delta R_2$$

$$\frac{\partial R}{\partial R_1}=\frac{R_2(R_1+R_2)-R_1 R_2}{(R_1+R_2)^2}=\frac{R_2^2}{(R_1+R_2)^2}$$

$$\frac{\partial R}{\partial R_2}=\frac{R_1^2}{(R_1+R_2)^2}$$

$$\Rightarrow \Delta R=\frac{R_2^2}{(R_1+R_2)^2}\Delta R_1+\frac{R_1^2}{(R_1+R_2)^2}\Delta R_2$$

$\Delta R_1 = 2200\times 0{,}15 = 330$

$\Delta R_2 = 120\times 0{,}15 = 18$

$\Delta R \approx 17{,}06896\,\Omega$

Finalement $\Delta R \leqslant 20\,\Omega$

donc $R=(110\pm 20)\,\Omega$

Calcul différentiel

$$\ln R=\ln(R_1 R_2)-\ln(R_1+R_2)=\ln R_1+\ln R_2-\ln(R_1+R_2)$$

donc $d(\ln R)=\frac{dR}{R}=d(\ln R_1)+d(\ln R_2)-d(\ln(R_1+R_2))$

$$\frac{dR}{R}=\frac{dR_1}{R_1}+\frac{dR_2}{R_2}-\frac{d(R_1+R_2)}{R_1+R_2}=\frac{dR_1}{R_1}+\frac{dR_2}{R_2}-\frac{dR_1}{R_1+R_2}-\frac{dR_2}{R_1+R_2}$$

$$\frac{dR}{R}=\left(\frac{1}{R_1}-\frac{1}{R_1+R_2}\right)dR_1+\left(\frac{1}{R_2}-\frac{1}{R_1+R_2}\right)dR_2=\frac{R_2}{R_1(R_1+R_2)}dR_1+\frac{R_1}{R_2(R_1+R_2)}dR_2$$

Ainsi $\frac{\Delta R}{R}=\frac{R_2}{R_1+R_2}\frac{\Delta R_1}{R_1}+\frac{R_1}{R_1+R_2}\frac{\Delta R_2}{R_2}$, et $\frac{\Delta R_1}{R_1}=\frac{\Delta R_2}{R_2}=0{,}15$

Donc $\frac{\Delta R}{R}=0{,}15$, donc ΔR = 17 Ω

CD14

1) $f'=\frac{n' R}{n'-n}=\frac{1{,}53\times 3{,}00}{1{,}53-1{,}00}=8{,}66037\,cm$

On ne peut pas savoir car on ne connaît pas l'incertitude.

2) $df'=\frac{\partial f'}{\partial n'}dn'+\frac{\partial f'}{\partial n}dn+\frac{\partial f'}{\partial R}dR=\left[\frac{R(n'-n)-n' R}{(n'-n)^2}\right]dn'+\left[-\frac{-n' R}{(n'-n)^2}\right]dn+\left[\frac{n'}{n'-n}\right]dR$

$$df'=\frac{-nR}{(n'-n)^2}dn'+\frac{n' R}{(n'-n)^2}dn+\frac{n'}{n'-n}dR$$

3) $\Delta f'=\left|\frac{-nR}{(n'-n)^2}\right|\Delta n'+\left|\frac{n' R}{(n'-n)^2}\right|\Delta n+\left|\frac{n'}{n'-n}\right|\Delta R$

Application numérique : $\Delta f' \approx 0{,}463606\,cm \rightarrow \Delta f' \leqslant 0{,}5\,cm$

Donc $f'=(8{,}7\pm 0{,}5)\,cm$

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CD15

1) t est fixé (ici t = 12 s)

$Q \approx 4{,}414553.10^{-3}$ C

$$\Delta Q = \left|\frac{\partial Q}{\partial C}\right|\Delta C + \left|\frac{\partial Q}{\partial E}\right|\Delta E + \left|\frac{\partial Q}{\partial R}\right|\Delta R$$

$$\Delta Q = [E\exp(\frac{-t}{RC}) + CE(\frac{-t}{R})(\frac{-1}{C^2})\exp(\frac{-t}{RC})]\Delta C + [Cexp(\frac{-t}{RC})]\Delta E + [CE(\frac{-t}{C})(\frac{-1}{R^2})\exp(\frac{-t}{RC})]\Delta R$$

$$\Delta Q = (E + \frac{Et}{RC})\exp(\frac{-t}{RC})\Delta C + Cexp(\frac{-t}{RC})\Delta E + (\frac{Et}{R^2})\exp(\frac{-t}{RC})\Delta R \approx 1{,}36115\cdot 10^{-3}\,C$$

$$\Delta Q \leqslant 2\cdot 10^{-3}\,C \quad \rightarrow \quad Q = (4 \pm 2)\cdot 10^{-3}\,C$$

$$\frac{\Delta Q}{Q} = (1 + \frac{t}{RC})\frac{\Delta C}{C} + \frac{\Delta E}{E} + \frac{t}{RC}\frac{\Delta R}{R} \approx 0{,}30833 \quad \rightarrow \quad \frac{\Delta Q}{Q} \approx 31 \quad \%$$

2) $\ln(Q) = \ln(CE\exp(\frac{-t}{RC})) = \ln(C) + \ln(E) - \frac{t}{RC}$

$$d(\ln(Q)) = \frac{dQ}{Q} = \frac{dC}{C} + \frac{dE}{E} - d(\frac{t}{RC}) = \frac{dC}{C} + \frac{dE}{E} - t\,d(\frac{1}{R}\frac{1}{C}) = \frac{dC}{C} + \frac{dE}{E} - t[\frac{1}{R}d(\frac{1}{C}) + \frac{1}{C}d(\frac{1}{R})]$$

$$\frac{dQ}{Q} = \frac{dC}{C} + \frac{dE}{E} - t[\frac{1}{R}(\frac{-1}{C^2}dC) + \frac{1}{C}(\frac{-1}{R^2}dR)] = \frac{dC}{C} + \frac{dE}{E} + \frac{t}{RC^2}dC + \frac{t}{CR^2}dR$$

$$\frac{dQ}{Q} = (1 + \frac{t}{RC})\frac{dC}{C} + \frac{dE}{E} + \frac{t}{RC}\frac{dR}{R}$$

D'où $\frac{\Delta Q}{Q} = (1 + \frac{t}{RC})\frac{\Delta C}{C} + \frac{\Delta E}{E} + \frac{t}{RC}\frac{\Delta R}{R}$

CD16

ATTENTION : convertir les angles en radians (sinon, pb dans la suite)

$n \approx 1{,}4979114$

$$dn = \frac{\partial n}{\partial A}dA + \frac{\partial n}{\partial D}dD = \frac{\frac{1}{2}\cos(\frac{A+D}{2})\sin(\frac{A}{2}) - \frac{1}{2}\cos(\frac{A}{2})\sin(\frac{A+D}{2})}{\sin^2(\frac{A}{2})}dA + \frac{\frac{1}{2}\cos(\frac{A+D}{2})}{\sin(\frac{A}{2})}dD$$

$$dn = \frac{1}{2}\frac{\sin(\frac{A}{2} - (\frac{A+D}{2}))}{\sin^2(\frac{A}{2})}dA + \frac{1}{2}\frac{\cos(\frac{A+D}{2})}{\sin(\frac{A}{2})}dD = -\frac{\sin(\frac{D}{2})}{2\sin^2(\frac{A}{2})}dA + \frac{\cos(\frac{A+D}{2})}{2\sin(\frac{A}{2})}dD$$

Ainsi, on a : $\Delta n = \left|\frac{-\sin(\frac{D}{2})}{2\sin^2(\frac{A}{2})}\right|\Delta A + \left|\frac{\cos(\frac{A+D}{2})}{2\sin(\frac{A}{2})}\right|\Delta D$

Donc $\Delta n = 0{,}015 \rightarrow \Delta n \leq 0{,}02$, donc $n = (1{,}50 \pm 0{,}02)$

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